Koen Essers • Master Internship • October 29, 2026

Notes Week 34

a. Bug Report

```
!! @param ACMI An array of the masses in AMU of those species.
ACMI(relevant_species) = chem_isos% W(chem_id(j))
A(relevant_species) = chem_isos% Z_plus_N(chem_id(j))


chem_isos% Z_plus_N = chem_isos% Z + chem_isos% N
chem_isos% W = chem_isos% Z_plus_N + chem_isos% mass_excess*(mev_to_ergs/amu/(clight*clight))
```

b. MESA mailing list

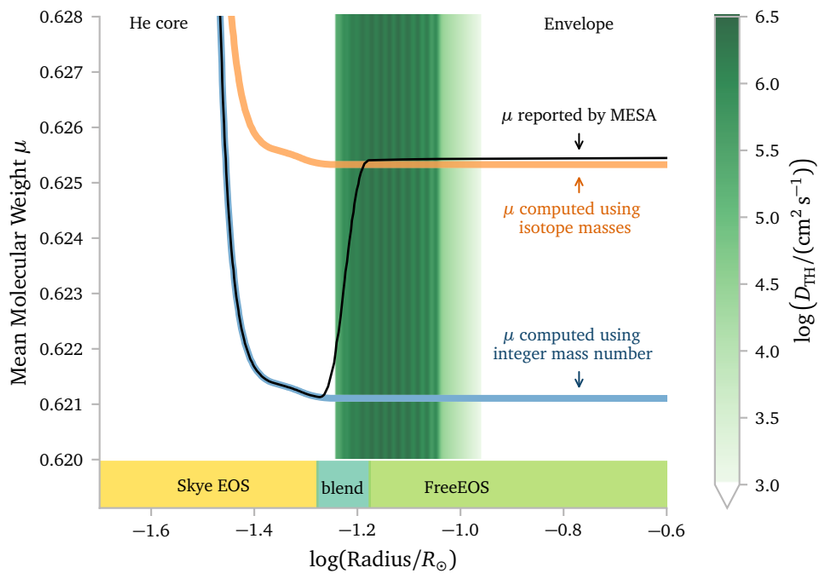
Hi everyone,

I noticed something while inspecting mean molecular weight profiles in MESA and I was wondering if I am observing the expected behaviour.

Skye EOS and FreeEOS appear to use different approaches to calculate the mean molecular weight μ. Skye uses integer isotope mass number (A_i), whereas FreeEOS appears to use its own table of elemental atomic weights.

These different approaches cause discrepancies in the mean molecular weight reported by the two different EOS tables. To show this, I ran the `hb_2M` test suite with thermohaline mixing enabled using mesa-r23.05.1. The MESA reported μ changes sharply across the EOS blending region, transitioning between the values obtained using these two approaches. During the RGB, the transition between the two tables occurs in a radiative region and appears to trigger thermohaline mixing.

The figure below shows the μ profile in the RGB-phase of the `hb_2M` model, together with μ values independently computed with the two approaches assuming complete ionization, and the thermohaline mixing diffusion coefficient.



Is this difference between the two μ calculations something that is already known? And is the resulting molecular weight gradient, and the associated thermohaline mixing expected when blending Skye and FreeEOS?

Thanks!

Best regards,

Koen

c. Summarizing my Work

I will first look at what I think is most relevant in each of my weekly notes.

2 – 3. This week contains a lot of relevant MESA parameters that I use, and *why* I use them. Other than that, there are not really any relevant results.

4. This week shows my first attempts of RLOF mass transfer with MESA. I also check some dredge-up properties of Rees et al. (2024)-like TPAGB models. There are no relevant *new* results.

5. I test different wind prescriptions for single stars. This is relevant for the methods section of my final thesis, but does not contain *new* results. I also compare different opacity tables. Similarly, this is important for the methods, but does not show something *new*.

   I also implement the quasi-adiabatic mass loss limit from Temmink et al. (2023). Again something for the methods. This is also the first week where I try some binary models with RLOF during the TPAGB phase of the donor. Not really any interesting results though, since these models still miss wind-mass transfer and tides.

6. Tried to figure out why models with low R and q fail. Did not really succeed. Contains a good description of ÆSOPUS tables for methods / introduction.

7. Contains detailed information of the overshooting parameters and comparisons with other methods. Also, MesaGrid class version 1. Not very relevant any more.

8. Contains the first implementation of the tides. This version does still contain some bugs but it is important for the methods section of my thesis. It also contains the first implementation of a model selector, something that I update many times later on. This is also interesting for the methods section of my thesis.

9. Fixes to tides, also the first implementation of different angular momentum losses due to winds, including the Saladino et al. (2018) winds. Shows detailed derivations as well for the angular momentum terms.

10. Starts with a nice figure showing the effects of the tides and wind angular momentum prescriptions for the Roche lobe evolution. Contains a lot of validations for the wind and tides, but this is not the complete and (hopefully) bug free setup yet. It does very nicely show how the individual parts of the prescriptions evolve over time. Also notes for the first time the difference between the two wind velocity assumptions and shows how they affect in particular Saladino et al. (2018).

11. Constant wind velocity of 15 km/s for everything in the model. This also contains important "last" tests for the winds + tides prescription. Shows full consistency with `evolve.py` code. Also shows some examples of early grid generation code and how I make sure the mesa models use the correct starting values. Pretty important for methods, not so much for results.

12. Lots of Soberman et al. (1997) analytic work. Not really any relevant results, but certainly helped with building intuition and understanding. Shows that Soberman et al. (1997) predictions of TPAGB-like mass transfer with modest δ values can already produce small orbits on the orbit of a couple hundred days. Also shows for the first time explicitly that small differences in initial binary separation can quite drastically change the outcome.

13. Implementation of proper circumbinary torus mass loss and the first grid showing some results. Also the first time encountering the Darwin instability but no nice plots yet. From this point onwards, the basic MESA models do not change a lot anymore, apart from additions that are required for abundances.

14. This week I show in more detail the Darwin instability. This is also the week that I actually enable the fully conservative mass transfer that Temmink et al. (2025) talks about.

15 – 16. Final model saving code. It now saves a single model at the start of a new thermal pulse with information on the maximum radius that was reached in the time before it. This week I also *observed* the sudden mixing that sometimes occurs in the other layers of the star, which are not convective, but I had not yet *understood* what I was seeing. I also look at the eccentricity and find that a lot of the parameter space is *expected* to be circularized if you *assume* Hut (1981) style tides. However, even with Hut (1981) style tides, there are still some models that can maintain their eccentricity at the start of RLOF.

17. Here I tried to get some semi-analytic bounds for relevant mass-transfer parameters β, γ and δ. It is based on the criterion that the radius of the star cannot ever become larger than the semi-major axis. This is somewhat interesting, *but* I think it is definitely not a major finding.

18. This week I compared the MESA internal tides with my own implementation. They produce slightly different results, but I opt to use my own method because it works well with the wind mass transfer prescription. The MESA tides does not work well with mass losses without having to implement a lot of new custom code.

19. This week I did some more analytic work. I tried to see the effects of q, δ and β. The most important result is that isotropic re-emission is *much* less important for the change in binary separation than circumbinary disk mass loss. This means isotropic re-emission can be "used" to tune the amount of accreted mass, while δ must be tuned to produce realistic final periods.

    Then I looked into the radial response of mass transfer for different δ values. I find that stronger δ values can cause a sort of delayed increase in the radius when a lot the mass is lost.

    In the HR diagram, this looks like a loop going from the right down, then moving to up and left to the Post-AGB constant luminosity track.

20. This week, I first tried to simulate a single star with a lower metallicity, based on metallicity data from observations of barium giants. Later, I change this metallicity to a more appropriate value for barium dwarfs. We see that a lower metallicity slightly alters the evolution of the star, but it is most important for the abundances.

    I also implement a very important fix to my angular momentum loss prescription to also take into account angular momentum lost from the accretor.

    I also implement a better stopping condition, making sure the models have finished RLOF. Then I also implement a profile saving method that saves profiles throughout mass loss.

21. Lower metallicity stars have larger dredge-up efficiencies, in line with literature (Herwig, 2005; Karakas et al., 2002). Lower metallicity stars also have larger C/O ratios, also in line with literature.

    I also compare my MESA models with Monash and NuGrid results without success. My later attemps are much better.

    I show that it is the mass transfer rate that determines how the star responds radially to mass loss. Large mass loss rates can cause the "loop" in the HR diagram, which is a hint at some not super stable mass loss.

    I also *show* with the MESA models that δ is much more important for the final period than β, and that a large δ leads to larger mass transfer rates as well.

    Lastly, I look at the metallicity of dwarfs.

22. I show that the amount of dredge-up for low mass (1 to 3 $M_{\odot}$) increases with mass in the MESA models, just like in the Monash Karakas & Lugaro (2016) models.

    These weeks, I also update the grid generator code and `MesaGrid` class to create and analyze the MESA grids more effectively.

    Here, I also simulate a bulk of the single star models.

23. I use the masses of observed barium dwarfs in combination with the simple binary code to find suitable binary candidates for the creation of barium dwarfs.

    I find that it is difficult to create barium dwarfs with masses that agree with observations using large donor stars, simply because of the mass ratio. It is also important to interact later on during the TPAGB, because only then has the star had a third dredge-up episode to enrich the envelope.

24. The MESA binary models have *a lot of* convergence problems. I developed a python pipeline that tries to fix these convergence problems, which is very useful.

25. Made sure my post-processed $M_{\rm DUP}$ calculation agrees well with the internal MESA dredge-up calculation which does not track the actual mass. It agrees well.

    Also wrote some code to allow the MESA models to evolve past difficult points in their evolution.

    Then I looked at delayed Thermal pulses with happen on the post-AGB or even the white dwarf sequence.

    Also fixed a bug that caused the models to use incorrect starting conditions.

26. This week I got the intershell abundances from Zara. I implemented the first version of my intershell and envelope abundance estimations. Later on I refine this significantly.

    I show here as well that s-process elements only start occuring *after* the first thermal pulse with significant TDU.

27. I compare MESA with the different Monash models from Karakas & Lugaro (2016). The differences I find are the same as described by Rees et al. (2024), which is *good*, because I use the Rees et al. (2024) inlists, just modified a bit.

    I also finalize the intershell abundance estimation method. I end up using the TP + offset method, which produces the most natural and consistent interpolation.

    I also look at some yields. I find that there are some considerable differences between my models and the Karakas & Lugaro (2016) models.

28. I look at the MESA abundances and compare them with the abundances I get after interpolating the Karakas & Lugaro (2016) models.

    I find that the Monash models and my models predict significantly different abundances, both in the envelope *and* in the intershell for important elements like carbon and oxygen.

    This is really a consequence of the different intershell abundances. If I use my post-processing code to predict the envelope abundances with MESA intershell abundances, I obtain the MESA envelope abundances.

    I fix a crucial bug, where I used the wrong metallicity, and reanalyze the yields. I now find that they are very much in line with Karakas & Lugaro (2016) yields. This is a nice consistency check. The same is true for the intershell mass fractions and the envelope mass fractions.

29. Here I simulate a bunch of main sequence stars.

    I also derive the equation to compute μ for a fully ionized mixture. I then use this to compute the average μ of accreted material for my mesa models.

    I also develop an algorithm to interpolate the μ profiles and then mix the accreted material with a certain μ value to get an estimate for the mixing mass due to the thermohaline instability.

    I find that the mixing mass is much more strongly dependent on the *amount* of accreted mass than the actual μ of the accreted mass.

    I also implement the Asplund et al. (2009) abundances to estimate the initial MS abundances.

30. Important bug fix for the amount of accreted material.

    Also started looking into differences between the MESA reported μ, Monash reported μ, and μ computed using abundances.

    I simulate the MS models again but with an initial helium abundance that aligns with the Karakas & Lugaro (2016).

    I implement spectroscopic abundances into my post-processing calculations and look at the effects of ε, $M_{\rm TPAGB}$ and q.

    I also simulate a broad R grid, which shows quite nicely that the initial Roche lobe radius is very important for possibility of producing a barium star.

    Here I also finally see why a lot of the low q and low R models fail. They move to unstable configurations with increasing temperatures and radii.

31. I try to *compute* μ but I fail to reproduce the MESA value.

    I look into the failing models and clearly see that the minimum entropy in the profile *rises* during the occurence of the instability. This is a clear hint that the model is *not* in thermal equilibrium and the critial mass-loss rate from Temmink et al. (2023) is higher than the actual critical mass loss rate.

    The HR diagram also clearly shows a departure to unstable stellar conditions.

    I also use the post-processing code in combination with the simple binary code to predict which binary configurations are interesting to simulate with MESA.

    I look into the effects of q, ϵ and $M_{\rm TPAGB}$.

    A particularly important result is that it is quite easy to create barium stars with enough accreted material, even with very low RLOF accretion efficiencies. For the abundances of the accretor, it is very important that the donor has experienced significant dredge up. Without it, it is impossible to create a barium star. This places a lower limit on the initial Roche lobe radius for every initial mass.

32. Here I define my large grid and start simulating.

    Here I also figure out that the Skye EOS (Jermyn et al., 2021) and FreeEOS use a different way to compute the mean molecular weight μ. This explains the weird dips.

    Then I compare some accurate MESA accretion models with the simple binary model with some assumptions. I find that overall, they agree quite well, *however*, most of the assumptions are not actually true and differences between *MESA* accretion post-processing and the simple binary post-processing can all be explained by the assumptions that the simple-binary makes.

    I make some scatter plots of the final periods and the [s/Fe] abundances. This clearly shows a trend. Short period binaries interact *before* dredge-up has enriched the star and therefore we cannot produce a barium star. Then there is quite a broad period range where various abundances and periods can exist. Then, for very large initial Roche lobe radii, the models all move to very large periods with relatively large abundances. It also seems like it is possible to create barium stars with wind alone up to quite large periods.

    It is clear as well that very low RLOF mass transfer accretion can easily produce barium stars. We also see that very low RLOF accretion is already enough to produce barium stars in the region where the is no[1] wind.

    I also find that for the 1.4 $M_\odot$ models, (semi) RLOF can occur without large overfill ratios for much of the models that I evolved. In some cases, semi-RLOF happens during the last interpulse which produces

    However, as δ increases, we see the overfill ratio increase as well. The same is true for decreasing q.

    Interactions during the last interpulse period seems to be quite interesting as here it seems to be possible to accrete material with *high* s-process abundances, with low overfill ratios, and the possibility to shrink the orbit.

33. I show that the artificial μ gradient introduces thermohaline mixing.

    It seems like low envelope masses during interpulses provide ideal opportunities for mass transfer with high accretion efficiencies, low overfill ratios, but still relatively short periods. The 1.4 $M_\odot$ model has a particularly low envelope mass before the final thermal pulse. Models that interact during this last interpulse are relatively stable during RLOF

    I look for other models with low envelope masses during the last thermal pulse to test this hypothesis and find that this seems to hold for other masses too.

    The 2.4 $M_\odot$ models have low overfill ratios and low envelope masses, while the 2.2 $M_\odot$ and 2.6 $M_\odot$ models have larger envelope masses *and* higher overfill ratios.

    Models that interact during the interpulse reach significantly lower peak mass loss rates. The RLOF mass loss rate is much more comparable to the wind mass loss rate.

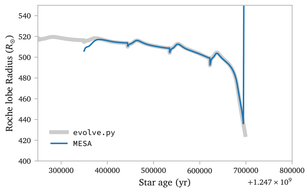
    There is a quite strong negative correlation between the final period and the overfill ratio. This means it is very difficult to produce low period binaries without large overfill ratios.

1 At least very little

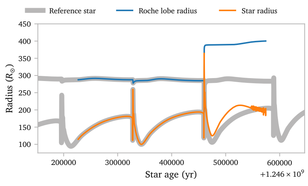
d. Results

Below I repeat important results with some preliminary conclusions.

d.1 Wind and Tides

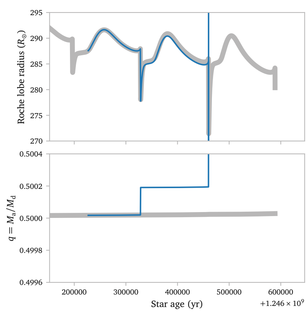


This plot shows the oscillations due to tides + stellar evolution and the sharp decrease in roche lobe radius due to strong winds.



In the weak wind regime, only the tides + stellar evolution affect the Roche lobe radius. Overal, we see a small decrease with time due to the increase in stellar radius, but it is much more subtle than in the strong wind regime. The MESA binary models agree well with the simple binary.

Below I plot a zoom



The Roche lobe radius is slightly smaller after the pulse, due to RLOF mass transfer.

I am sure I can make plots with nice annotations that more clearly show the effects. I think these results are interesting as they really show how systems with a TPAGB donor are driven into RLOF

d.2 Binary configurations

I varied donor mass, accretor mass and initial separation for my models. All models have eccentricity 0, which is an assumption that probably is not valid. However, it is also very uncertain how one *should* treat the eccentricities, let alone mass transfer in eccentric binaries.

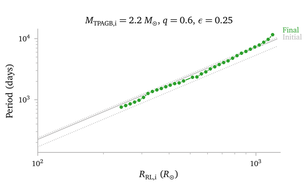
I also parameterise the binary systems using the donor mass, mass ratio, and initial Roche-lobe radius, rather than the individual stellar masses and initial orbital separation. These parameters are directly related to the latter but provide a more physically intuitive description of the mass-transfer configuration.

First of all, I find that many configurations are stable if we only consider the thermal response of the donor, in line with results from Temmink et al. (2023).

I find the results of mass transfer are strongly dependent on the thermal pulse [2] during which it occurs. The exact thermal pulse evolution is very strongly dependent on the initial mass. This is also not trivially parametrisable. It is especially important how much mass is left in the envelope during the last interpulse and thermal pulse.

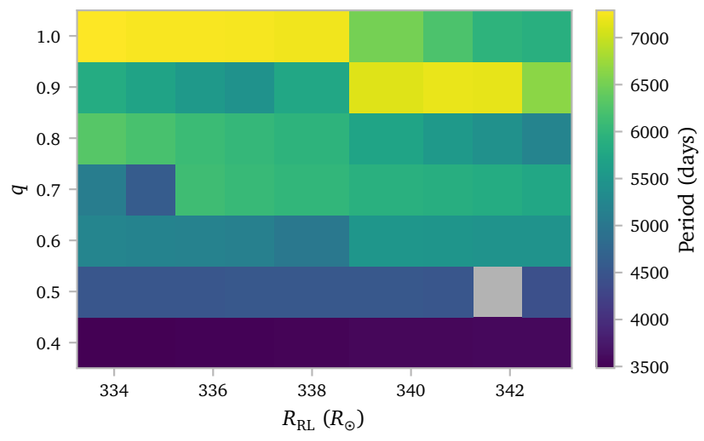
The Roche lobe radius determines, in many cases, much more strongly than the mass ratio, during which thermal pulse the donor star will undergo RLOF

2 or interpulse



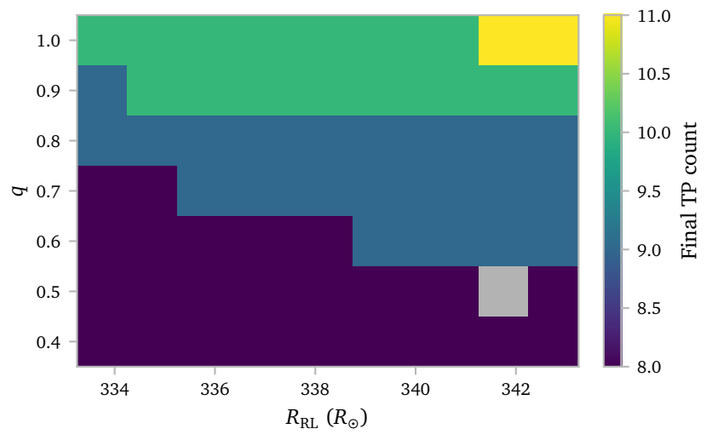
Here I show a broad range of models with $q = 0.6$, $M_{\rm TPAGB} = 2.2\,M_\odot$ and $\varepsilon = 0.25$. I also plot the initial period of the binaries, as well as the initial periods if we use $q = 0.1$ and $q = 1$.

We see that the final period is quite closely related to the initial Roche lobe radius. The initial mass ratio q and accretion efficiency parameters primarily shift the line up or down.



In some cases, the differences are quite drastic. Nearly identical starting orbits can end with very different separations. I think it is important to try to understand where these differences come from.

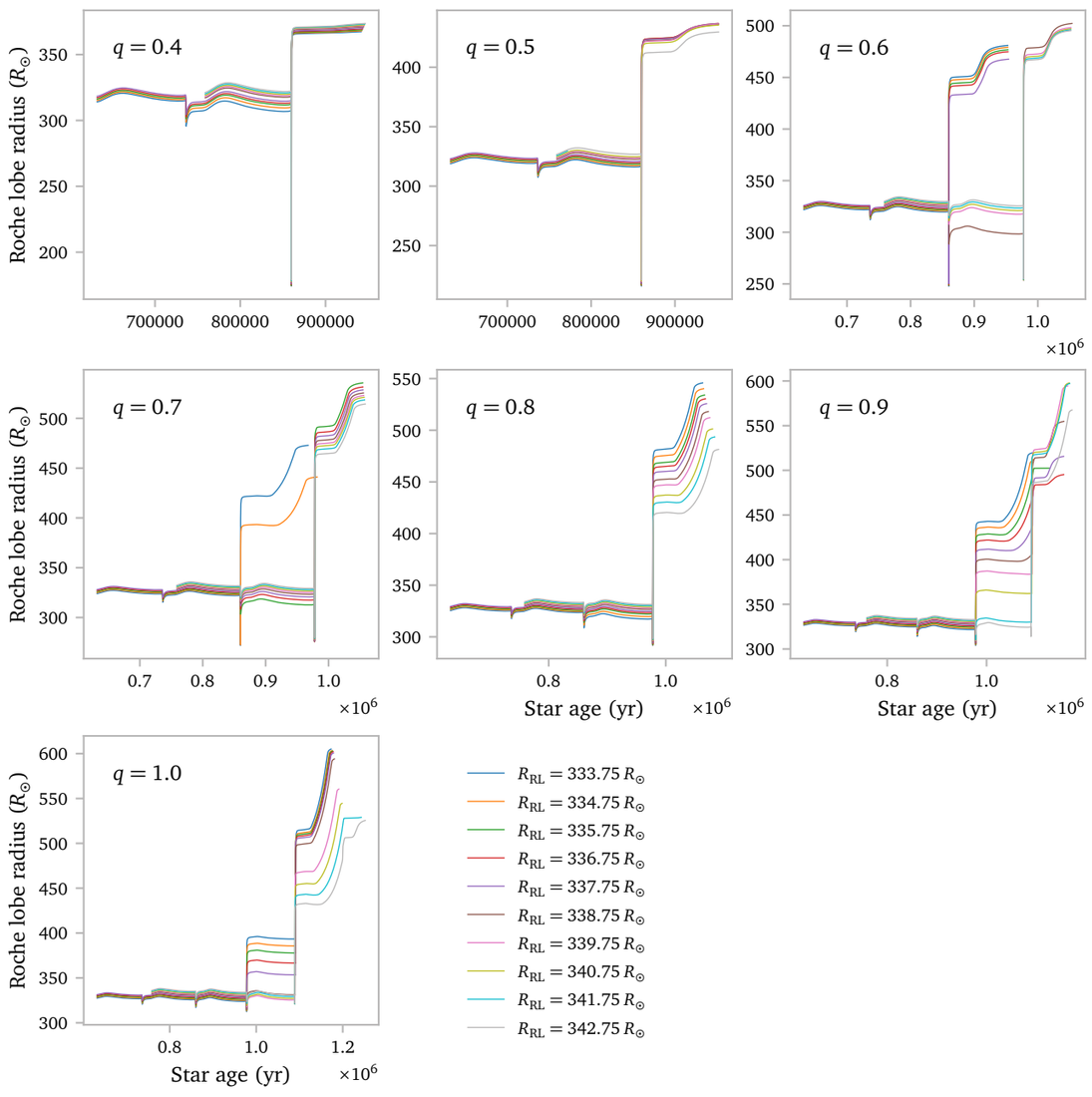
Presumably, this is due to interactions during a different thermal pulse. To check, we can look at the final TP count of the binary systems.

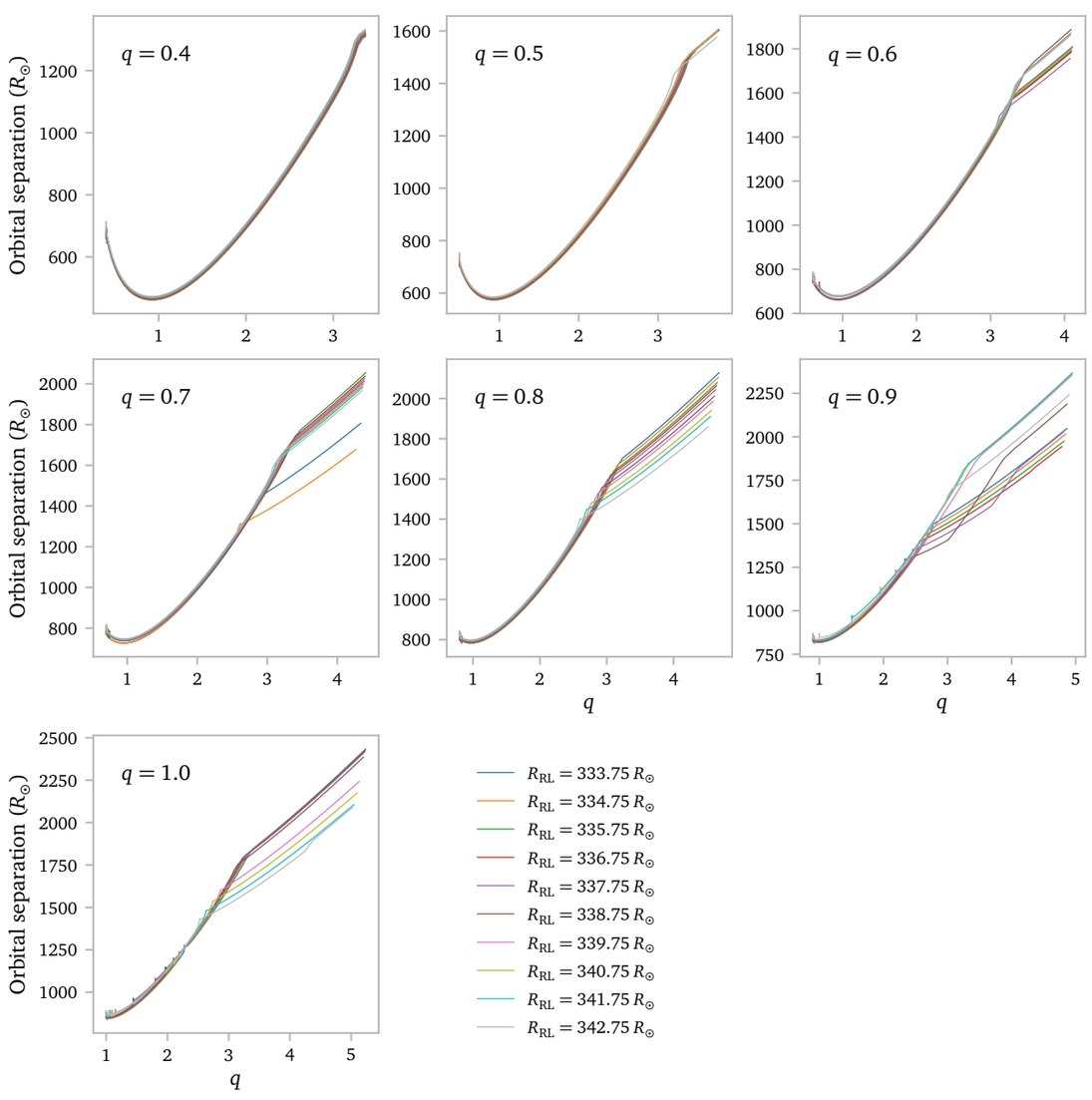


Although not a perfect match (for example, $q = 0.9$ shows nearly all systems experiencing the same number of pulses while clear differences in final periods exist) there does seem to be some relation between the thermal pulse count and the jumps in final period.

The plots below show the evolution of *all* models of this fine grid. They are split by q values. This makes it easier to compare. These are thus able to show the time evolution of the models of a single column in the plots above.

I start with a plot that shows the Roche lobe radius as a function of time.

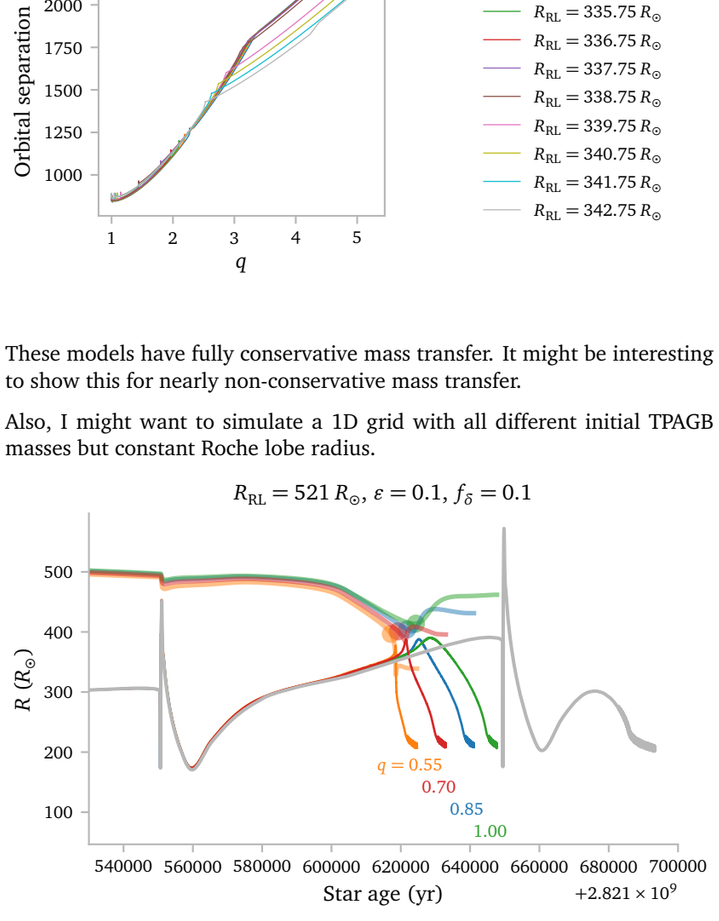




These models have fully conservative mass transfer. It might be interesting to show this for nearly non-conservative mass transfer.

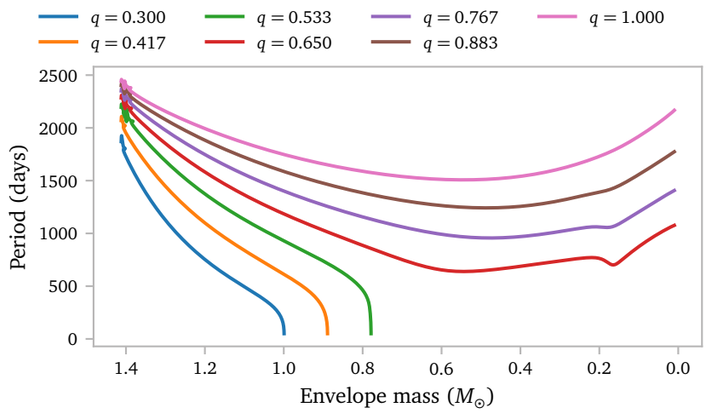
Also, I might want to simulate a 1D grid with all different initial TPAGB masses but constant Roche lobe radius.

$R_{\mathrm{RL}} = 521\,R_\odot$, $\varepsilon = 0.1$, $f_\delta = 0.1$



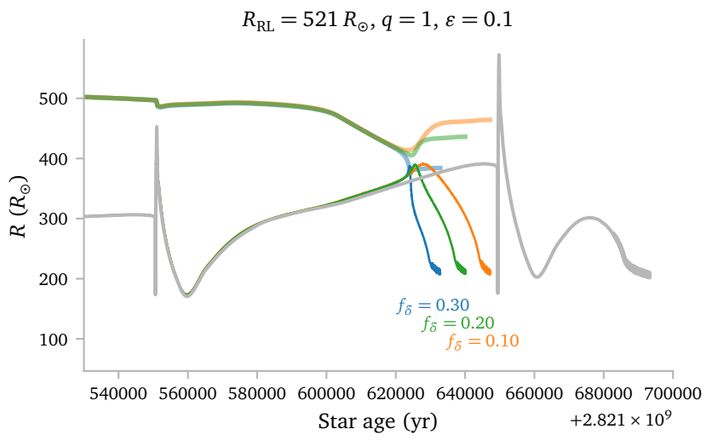
Apparently, q, has quite a large influence on *if* the model reaches significant overfill ratios. This makes sense considering what we have seen much earlier, where the models with the most unequal mass ratios also have the most extreme RLOF episodes, the most instabilities and the possibility to spiral in due to tides.

Low q models can reach a Darwin-instability like inspiral.



d.3 Accretion efficiencies

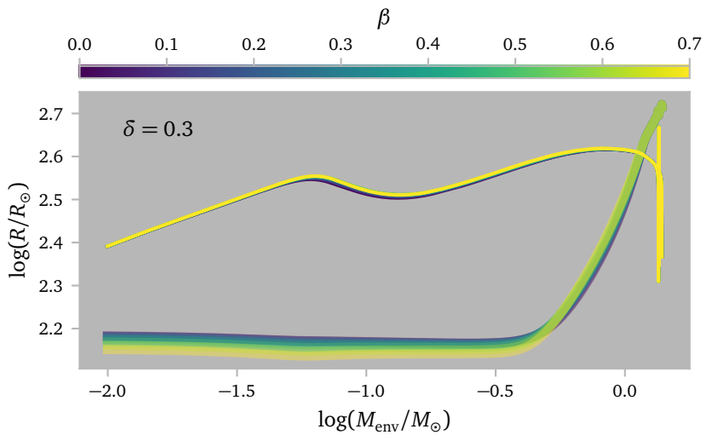
The mass transfer efficiency ε and the different values for isotropic re-emission and circumbinary coroting toroid mass loss have little influence on *when* systems undergo RLOF. However, they have very important consequences on *how* RLOF occurs.



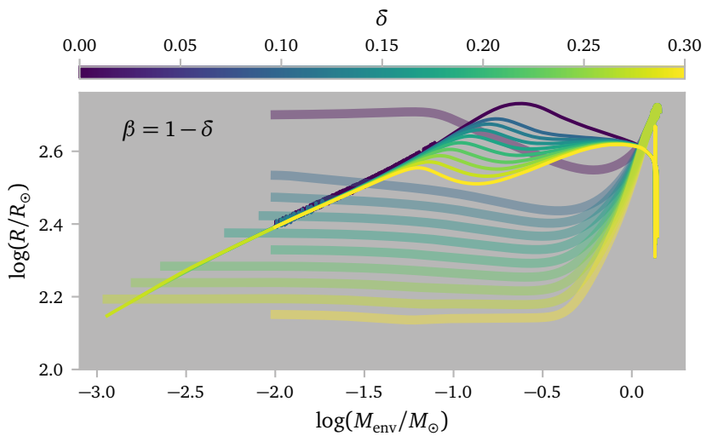
The effect of this is similar to what we see for different q values prior during RLOF, which is what we expect since *both* of these parameters alter the severity of RLOF. However, an important difference between the two is that the mass ratio *also* affects the winds, while δ does *not*. Therefore, prior to RLOF, the models with different δ will behave more similarly than the models with different q.

Larger δ primarily affects the overfill ratio, the duration of RLOF and the final period. We obtain much shorter final periods with larger δ, but this also makes RLOF less stable.

β however, is way less important for RLOF, but it does still affect how much mass can be accreted. This gives β a nice place in our prescription to tune the accretion efficiency without affecting very significantly the orbital response to RLOF.



The effect on the star radius is minimal, while a larger β tends to result in somewhat smaller final Roche lobe radii. This corresponds to shorter periods too.



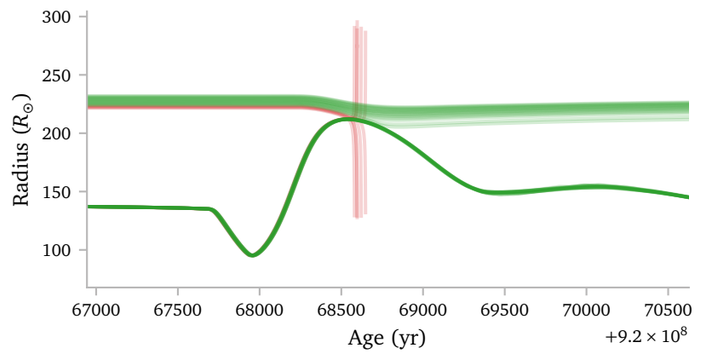
Clearly, changing δ has a much larger effect than β.

Some models stop because the binary spirals in. I have not plotted these.

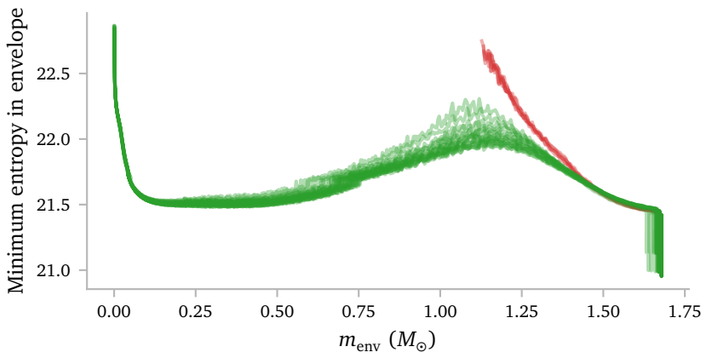
Some models only ran until $M_{\rm env} = 0.01 M_\odot$ because I forgot to remove the MESA default envelope mass stopping condition.

d.4 Thermal Response

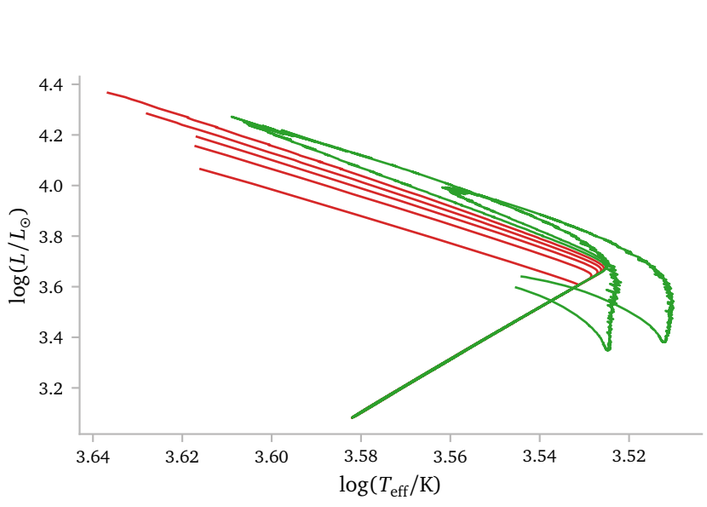
I think it might also be interesting to talk about the donors that are unstable to RLOF. Although they should be stable according the criterion from Temmink et al. (2023), I find that the minimum entropy in the envelope of these models increases, as they move to higher temperatures and radii during RLOF. MESA is unable to simulate these models.



This first plot simply shows the Roche lobe radius and the radius of the stars as a function of time. We can see that all of the failing models interact during 1 thermal pulse, while all of the succeeding models interact 1 thermal pulse later.



The bottom panel again shows the minimum entropy as a function of envelope mass. This time the resolution is much better because I compute this for every history step. We can now even more clearly see that the failing models tend to get much larger minimum entropies when losing their envelope.



d.5 Accretor Abundances

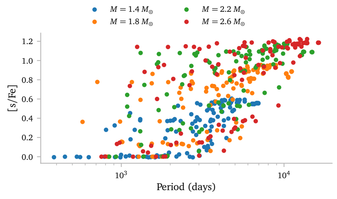
It seems like it is possible to create barium dwarfs with very little RLOF accretion, and wind accretion alone. This means the accretion efficiency of RLOF does not need to be high, which is expected as well due to the large overfill ratios and large mass-transfer rates.

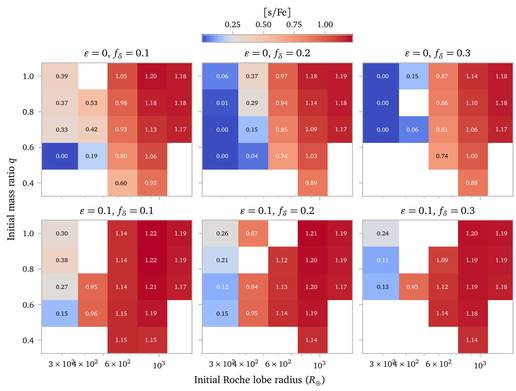
Quite unsurprisingly, we find that higher mass TPAGB-stars can produce more enriched accretors. We also find that it barium stars can only be created if the envelope of the TPAGB-star has already obtained enough dredged-up mass due to third dredge-up episodes.

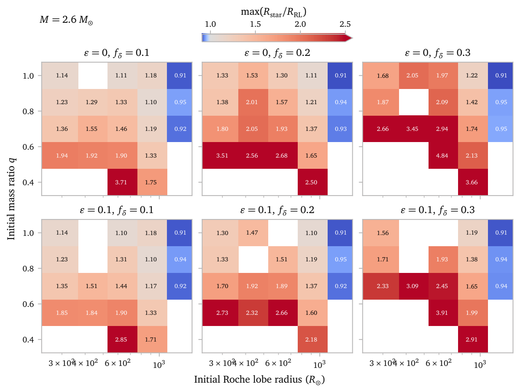
This is kind of unfortunate, as it requires larger initial Roche lobe radii, and thus also likely larger final periods. Thus it is difficult to produce very low period, s-process enhanced TPAGB stars.

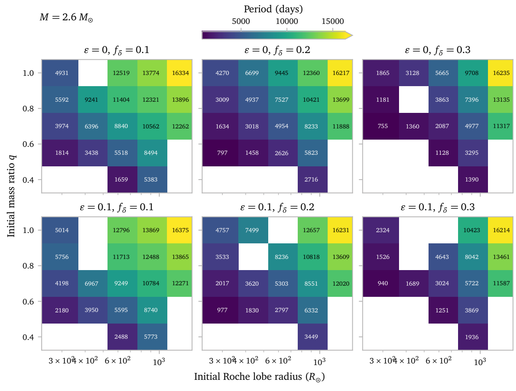
I think the plot below shows this well.

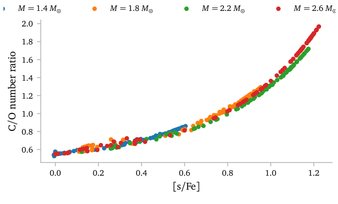
e. 2.6 Solar mass grid



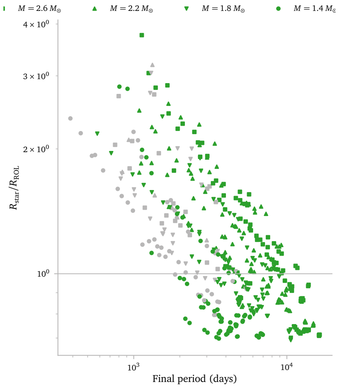






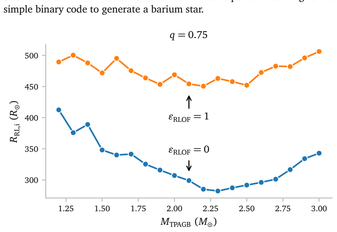


We see a very strong correlation between the [s/Fe] values and the C/O number ratios. This is quite different from observations, where this correlation is *much* less strong.



f. Additional Plots

Shows the minimum Roche lobe radius that is required according to the simple binary code to generate a barium star.



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